



UNITED GROUP OF INSTITUTIONS



**DEPARTMENT OF
COMPUTER SCIENCE & ENGINEERING /
COMPUTER SCIENCE & ENGINEERING (ARTIFICIAL
INTELLIGENCE & MACHINE LEARNING) /
COMPUTER SCIENCE & ENGINEERING (DATA
SCIENCE)**

**ASSIGNMENT BOOKLET
BACHELOR OF TECHNOLOGY
2ND YEAR (3rd Semester)**

AS PER AKTU CURRICULUM

SESSION -2025-2026

Assignment- 1

Data Structure (BCS-301)

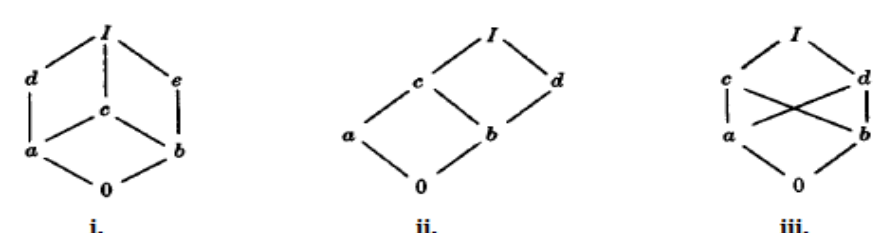
Unit- 1

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1.	Differentiate between overflow and underflow condition in a linked list.	CO1	K2
2.	How can you represent a sparse matrix in memory?	CO1	K2
3.	What are the various asymptotic notations?	CO1	K1
	Questions (Long Answer Questions)		
4.	Assume that the declaration of multi-dimensional arrays X and Y to be, X (-2:2, 2:22) and Y (1:8, -5:5, -10:5) [2022 2023] (i) Find the length of each dimension and number of elements in X and Y. [2022-2023] (ii) Find the address of element Y (2, 2, 3), assuming Base address of Y = 400 and each element occupies 4 memory locations.	CO1	K3
5.	Write an algorithm to insert a node at the end in a Circular linked list.	CO1	K3
6.	What do you understand by time space trade off? Define the various asymptotic notations. Derive the O- notation for linear search.	CO1	K2
7.	Write a program in C to delete a specific element in single linked list. Double linked list takes more space than single linked list for storing an extra address. Under what condition, could a double linked list more beneficial than single linked list.	CO1	K3
8.	Discuss the representation of polynomial of single variable using linked list. Write 'C' functions to add two such polynomials represented by linked list.	CO1	K2
9.	What is doubly linked list? What are its applications? Explain how an element can be deleted from doubly linked list using C program.	CO1	K2
10.	Discuss array and linked representation of queue data structure. What is dequeue?	CO1	K2

Discrete Structure and Theory of Logic (BCS-303) Unit- 1

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
Questions (Short Answer Questions)			
1	Define empty set and Power Set with example.	CO1	K2
2	Let $A = \{1, 2, 3, 4, 5, 6\}$, construct pictorial description of the relation R on A defined as $R = \{(a, b) : (a-b)^2 \in A\}$.	CO1	K3
3.	Let $A = \{1, 2, 3, 4\}$, and R is a relation defined by "a divides b". Write R as a set of ordered pair, draw directed graph. Also find R^{-1} .	CO1	K3
Questions (Long Answer Questions)			
4.	Answer these questions for Lattice $(D_6, /)$. Where D_n is a set of all divisors of n . The $/$ is defined as a divides b . Is it a complemented Lattice? Is it a bounded lattice? Is it a complete lattice?	CO1	K3
5.	Show that a relation R defined on the set of real numbers as $(a, b)R(c, d)$ iff $a^2 + b^2 = c^2 + d^2$. Show that R is an equivalence relation.	CO1	K3
6.	Which of the partially ordered sets in Fig are lattices?  i. ii. iii.	CO1	K3
7.	Draw the Haase diagram of $[p(a, b, c), \leq]$, Find greatest element, least element, minimal element & maximal element. here $\leq$ is subset symbol	CO1	K3
8.	Define Complete and distributive lattice.	CO1	K2

Computer Organization and Architecture (BCS-302)

Unit- 1

Note:- Attempt all questions in brief.

S.No .	Questions	COs	Bloom's Taxonomy
1.	Describe why a computer is termed as a digital system.	CO1	K2
2.	Describe different types of busses and their relevance in computer organization.	CO1	K2
3.	Describe the differences between computer architecture and computer organization.	CO1	K2
4.	What is mean by bus arbitration? List various types of bus arbitration.	CO1	K2
5.	Draw the basic functional units of computer.	CO1	K2
6.	Describe the role of Multiplexer and Decoder.	CO1	K2
7.	Draw a diagram of bus system using MUX which has four registers of size 8-bits each.	CO1	K3
8.	Describe processor organization in detail.	CO1	K2
9.	Evaluate the arithmetic statement $X=A+B*[C*D+E*(F+G)]$ Using a stack organized computer with zero address operation instructions.	CO1	K3
10	Write short note on addressing mode.	CO1	K2

Universal Human Values (BVE-301)

Unit- 1

Note:- Attempt all questions in brief.

SECTION –A (SHORT ANSWER TYPE QUESTIONS)		CO	Blooms Taxonomy Level
1	A	What are the abbreviations given as SVDD, SSDD and SSSS signify?	CO1 K1
	B	What is the difference between ‘value’ and ‘skills? Do they complement each other?	CO1 K2
	C	What is the difference between human values and moral values?	CO1 K2
	D	What do you understand by the value of an entity?	CO1 K2
	E	What do you understand by the terms svatva, swatantrata and swarajya?	CO1 K1
SECTION –B (LONG ANSWER TYPE QUESTIONS)			
2	A	Discuss the need, basic guidelines, content, and process of value education.	CO1 K1
	B	There are many problems manifest today at the level of individual, family, society and the nature. Identify some of these problems humans suffer from.	CO1 K2
	C	What is happiness & Prosperity? Are they related to each other? Can happiness be obtained without prosperity? Explain.	CO1 K2
	D	What are the basic human aspirations? Explain the basic requirement for fulfillment of human aspirations?	CO1 K2
	E	What do you mean by your natural acceptance and experiential validation? Explain the process of Self Exploration.	CO1 K2

Technical Communication (BAS-301)

Unit- 1

Note:- Attempt all questions in brief.

S.No.	Questions	CO	Bloom's Taxonomy
1.	Write a detailed note on important features of technical communication.Or Discuss the process of communication with the help of diagram.	CO1	K2
2.	List differences 10 between general and technical communication.	CO 1	K1
3.	Explain horizontal / lateral (peer group) flow of communication. Why is it beneficial than other formal flows?	CO 1	K2
4.	Discuss the Intrapersonal barrier along with its different factors and ways to overcome it.	CO 1	K1
5.	Discuss the role of 7 C's of communication.	CO 1	K2
6.	What is code and content?	CO 1	K2
7.	What is Pronunciation? How does it affect the communication? OR Briefly answer the following : Vowel sound ,Consonant sound , Rising tone & Falling tone	CO 1	K1
8.	What is Intonation? Discuss all three types of Intonation in detail.	CO 1	K1
9.	Describe barriers to effective communication. How they are created and ways to overcome?	CO 1	K2
10.	Stimulus & response is extensively used by mass media for various purposes.Explain the statement with suitable examples.	CO 1	K2

Digital Electronics (BOE-310)

Unit- 1

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1.	Convert the followings in binary number system (i) $(765.245)_{10}$ (ii) $(34.654)_8$ (iii) $(332.13)_5$ (iv) $(543.66)_7$ (v) $(5D3.AC2)_{16}$.	CO1	K1
2.	Convert the following in decimal number system (i) $(1101.11)_2$ (ii) $(723.45)_8$ (iii) $(332.13)_5$ (iv) 543.66_7 (v) $(5D3.AC2)_{16}$	CO1	K1
3.	Find binary code of following gray codes (i) $(10110)_2$ (ii) $(111111111)_2$ (iii) $(111000)_2$	CO1	K1
	Questions (Long Answer Questions)		
4.	Find the gray code of followings: (i) $(10110)_2$ (ii) $(111111111)_2$ (iii) $(111000)_2$	CO1	K1
5.	A 7-bit hamming code is received as 1001101, check for the errors if any, and then correct the code if even parity scheme is used.	CO1	K3
6.	Minimize the function using K-map $Y = \sum m (1,2,3,9,11,12,13,14) + d (0,7,10)$	CO1	K2
7.	Minimize the function using K-map $F = \prod M (4,5,6,13,14,15)$	CO1	K2
8.	Minimize the following function using Q-M or Tabular method: $F(A, B, C, D) = \sum m (4,6,7,8,9,10,11,15)$	CO1	K3
9.	Draw the NOR-NOR implementation for $F = (A+B). (C+D). E$	CO1	K2
10.	Draw the logic diagram of Ex-OR & Ex-NOR gate using NAND gate only	CO1	K2

Mathematics-IV (BAS-303)

Unit- 1

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1.	Solve the PDE: $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$.	CO1	K2
2.	Solve the PDE: $2zx - px^2 - 2qxy + pq = 0$.	CO1	K2
3.	Solve the PDE: $r + 5s + 6t = \frac{1}{y-2x}$ where the symbols have their usual meanings.	CO1	K1
	Questions (Long Answer Questions)		
4.	Solve the PDE: $r - 3s + 2t = e^{2x+3y} + \sin(x - 2y)$ where the symbols have their usual meanings.	CO1	K2,K3
5.	Solve the PDE: $r - 6s + 9t = 12x^2 + 36xy$ where the symbols have their usual meanings.	CO1	K1
6.	Solve the PDE: $r + s - 6t = y \cos x$ where the symbols have their usual meanings.	CO1	K2,K3
7.	Solve the PDE: $(D - D' - 1)(D - D' - 2)z = e^{3x-y} + x$.	CO1	K4
8.	Solve the PDE: $(D - D' - 1)(D - D' - 2)z = \sin(2x + 3y)$.	CO1	K2
9.	Solve the PDE: $D(D + D' - 1)(D + 3D' - 2)z = x^2 - 4xy + 2y^2$.	CO1	K4
10.	Solve the PDE $(D^2 - D'^2 - 3D + 3D')z = xy + e^{x+2y}$.	CO1	K2

Python Programming (BCC-302)

Unit- 1

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1.	What is a variable in Python? List any two basic operators in Python.	CO1	K1
2.	What is the difference between int and float data types?	CO1	K2
3.	Write a simple assignment statement in Python.	CO1	K2
4.	Which operator is used for exponentiation in Python?	CO1	K2
5	Give one example of a Python block.	CO1	K1
	Questions (Long Answer Questions)		
6.	Explain the concept of variables in Python with suitable examples.	CO1	K2
7.	Explain Python numeric data types (int, float, complex) with examples.	CO1	K2
8.	Write a Python program that takes two numbers as input and performs addition, subtraction, multiplication, and division.	CO1	K1
9.	Differentiate between assignment operators and comparison operators with examples	CO1	K2
10.	Discuss different types of Python operators with examples.	CO1	K3

Assignment- 2

Data Structure (BCS-301)

Unit- 2

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1	Draw a binary tree for the expression: $A * B - (C + D) * (P / Q)$.	CO2	K2
2	Give some applications of Stack.	CO2	K3
3.	Which data structure is used to perform recursion and why?	CO2	K3
	Questions (Long Answer Questions)		
4.	Write an algorithm to convert a valid arithmetic infix expression into an equivalent postfix expression. Trace your algorithm for the following infix expression. [2023 2024] $A + B * C - D / F$.	CO2	K3
5.	Write algorithm for push and pop operations in Stack. Transform the following expression into its equivalent postfix expression using stack: $A + (B * C - (D / E \wedge F) * G) * H$	CO2	K3
6.	A double ended Queue (deque) is a linear list in which additions may be made at either end. Obtain a data representation mapping a deque into one dimensional array, Write C function to add and delete elements from either end of deque.	CO2	K3
7.	Write a C program to calculate factorial of number using recursive and non-recursive functions.	CO2	K3
8.	What is circular Queue? Write a C code to insert an element in circular queue?	CO2	K2
9.	Write A C programme to implement stack using single linked list.	CO2	K3
10.	Design a method for keeping two stacks within a single linear array so that neither stack overflow until all the memory is used.	CO2	K3

Discrete Structure and Theory of Logic (BCS-303) Unit- 2

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
Questions (Short Answer Questions)			
1	Let $A = \{1,2,3\}$, $B = \{p,q\}$ and $C = \{a,b\}$. Let $f: A \rightarrow B$ is $f = \{(1,p), (2,p), (3,a)\}$ and $g: B \rightarrow C$ is given by $\{(p,b), (q,b)\}$. Find $g \circ f$ and show it pictorially.	CO2	K2
2	Write description about omega, theta and big O notations.	CO2	K3
3.	Write any three algebraic laws of Boolean algebra.	CO2	K3
Questions (Long Answer Questions)			
4.	List all possible functions from A to B, $A = \{a,b,c\}$, $B = \{0,1\}$. Also indicate in each case whether the function is Injective , Surjective or Bijective.	CO2	K3
5.	Simplify the following Boolean functions using K-map: (a) $F(x,y,z) = \Sigma(0,1,5,7)$ (b) $F(x,y,z) = \Sigma(1,2,3,6,7)$	CO2	K3
6.	Simplify using K Map $F(A,B,C,D) = \Sigma(m_0, m_1, m_2, m_4, m_5, m_6, m_8, m_9, m_{12}, m_{13}, m_{14})$	CO2	K3
7.	Draw the logic diagram to represent the Boolean expression: $(xy') + xyz$	CO2	K3
8.	Express $E(x, y, z) = xy + y'z$ into its complete sum-of-products form.	CO2	K2
9.	Investigate whether the function $f(x) = x^2 - 1$ is injective or not for $f: \mathbb{R} \rightarrow \mathbb{R}$.	CO2	K3
10.	Solve the following Boolean function using K-map: $F(A,B,C) = (1,2,5,7)$ and $D(0,4,6)$ using SOP.	CO2	K3

Computer Organization and Architecture (BCS-302)

Unit- 2

Note:- Attempt all questions in brief.

S.No.	Questions	COs	Bloom's Taxonomy
1.	Write the decimal number 35 and -35 in 2's sign magnitude format.	CO2	K2
2.	Register A holds a binary value of 10011110. What is the register value after the arithmetic shift right? Starting from the initial number 10011101, determine the register value after the arithmetic shift left, and state whether there is an overflow.	CO2	K2
3.	Design a 4-bit combinational incremental circuit using four full adder circuits.	CO2	K2
4.	Write the 4-bit bidirectional shift register with its function table.	CO2	K2
5.	Perform 2's complement subtraction of smaller number (101011) from larger number (111001).	CO2	K2
6.	A binary floating-point number has seven bits for biased exponent. The constant used for the bias is 64. I. List the biased representation of all exponent from -64 to +63. II. Show that after addition of two biased exponents, it is necessary to subtract 64 in order to have a biased exponent's sum. Show that after subtraction of two biased exponents, it is necessary to add 64 in order to have a biased exponent's difference.	CO2	K2
7.	Demonstrate floating-point number representation in computer, also give IEEE 754 standard 32-bit floating point number format of (12.21) ₁₀ justifying each step.	CO2	K3
8.	What is restoring method in division algorithm?	CO2	K2
9.	What is the role of Multiplexer and Decoder?	CO2	K3
10.	Describe Sequential arithmetic & logic unit (ALU) using proper diagrams with its function table	CO2	K2

Universal Human Values (BVE-301)

Unit- 2

Note:- Attempt all questions in brief.

SECTION –A (SHORT ANSWER TYPE QUESTIONS)			CO	Blooms Taxonomy Level
1	A	What do you mean by Sukh and Suvidha?	CO2	K1
	B	The needs of the body are quantitative. Illustrate.	CO2	K2
	C	What do you mean by Imagination?	CO2	K2
	D	“I am the seer, doer and enjoyer. The body is my instrument” – Explain.	CO2	K2
	E	Briefly explain the activities of Desire, Thought, and Expectation in the self with an example.	CO2	K1
SECTION –B (LONG ANSWER TYPE QUESTIONS)				
2	A	Distinguish between the needs of the Self and the needs of the Body.	CO2	K1
	B	What is the meaning of desire? How do we verify whether our desires are coming from sensations or preconditioning or natural acceptance?	CO2	K2
	C	What are the programs for ensuring the health of the body? Explain.	CO2	K2
	D	What do you mean by Sanyam? How does it ensure harmony with the body? Explain.	CO2	K2
	E	In what way can we say that the human body is a self-organized unit?	CO2	K2

Technical Communication (BAS-301)

Unit- 2

Note:- Attempt all questions in brief.

S.No.	Questions	CO	Bloom's Taxonomy
1.	Differentiate between CV and Résumé. Which one is better and why?	CO2	K2
2.	Draft a résumé for the post of software engineer in a reputed company. Invent necessary details.	CO2	K1
3.	What is Interview? Explain five types of Interviews.	CO2	K2
4.	Write a short note on communication and networking.	CO2	K1
5.	Define Profiling. Draft a profile highlighting skills and qualities to attract your employer.	CO2	K2
6.	Discuss the three phases of Interview process. Explain with reason how different skills are essential to combat these phases?	CO2	K2
7.	What are the objectives and salient features of Group Discussion?	CO2	K1
8.	What is an expert technical lecture? How can audio- visual aids be helpful in the delivery of the same?	CO2	K1
9.	What do you understand by SOP? What should be included in a SoP?	CO2	K2
10.	Explain the process of writing a conference paper. Discuss the structure, style and content required for effective presentation on stage	CO2	K2

Digital Electronics (BOE-310)

Unit- 2

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1.	Define the term combinational logic with an example.	CO2	K1
2.	How many 4×1 multiplexers are required to implement 64 ×1 multiplexer?	CO2	K1
3.	Implement a 4:1 multiplexer using 2:1 multiplexer.	CO2	K1
	Questions (Long Answer Questions)		
4.	Demultiplexer is decoder circuit with an additional enabling input. Do you agree with the above statement?	CO2	K1
5.	Design a full adder using NAND gates only.	CO2	K2
6.	Implement a full subtractor circuit using only NAND gates.	CO2	K2
7.	Using 4:1 multiplexer, implement the following function $F(A, B, C) = \sum m(0,2,3,5,7)$	CO2	K2
8.	Explain 4-bit magnitude comparator.	CO2	K2
9.	Design a BCD adder using 4-bit parallel adder.	CO2	K2
10.	Construct a full adder and implement the full adder with the help of half adders. Also implement the full adder with NAND gates only.	CO2	K2

Mathematics-IV (BAS-303)

Unit- 2

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1.	Classify the PDE $u_{tt} + tu_{xt} + xu_{xx} + 2u_t + u_x + 6u = 0$.	CO2	K2
2.	Show that the partial differential equation $z_{xx} + 2xz_{xy} + (1 - y^2)z_{yy} = 0$ is elliptic for all values of x and y in the region $x^2 + y^2 < 1$, parabolic on the boundary and hyperbolic outside this region.	CO2	K2
3.	Solve the PDE $z_x + z_{yy} = 0$ by the method of separation of variables given that $z(x, 0) = 0, z(x, \pi) = 0, z(0, y) = 4 \sin 3y$	CO2	K3
	Questions (Long Answer Questions)		
4.	Solve the PDE $u_{xx} = 2u + u_y$ by using the method of separation of variables subject to conditions $u = 0$ and $u_x = e^{-3y}$ when $x = 0$ for all values of y .	CO2	K3
5.	An elastic string of length l which is fastened at its ends $x = 0$ & $x = l$ is picked up at its center point $x = \left(\frac{l}{2}\right)$ up to a height $\left(\frac{l}{2}\right)$ and released from rest. Find displacement of the string at any instant of time.	CO2	K4
6.	A string is stretched and fastened to two points distance l apart. Find the displacement $y(x, t)$ at any point at a distance x from one end at a time t , given that $y(x, 0) = A \sin \frac{2\pi x}{l}$.	CO2	K3
7.	The heat flow in a bar of 10 cm of homogeneous material is governed by PDE $u_t = c^2 u_{xx}$. The ends of bar are kept at zero temperature and initial temperature is given by $f(x) = 10 - x$. Find temperature distribution in bar.	CO2	K3
8.	Find the temperature in a bar of length 2 whose ends are kept at zero and lateral surface insulated if the initial temperature is $\sin \frac{\pi x}{2} + 3 \sin \frac{5\pi x}{2}$.	CO2	K3
9.	A semicircular plate of radius a has its circumference kept at temperature $u(a, \theta) = k\theta(\pi - \theta)$ while the boundary diameter is kept at 0°C . Find the steady state temperature distribution $u(r, \theta)$ of the plate assuming the lateral surfaces of the plate to be insulated.	CO2	K4
10.	Solve the partial differential equation $\frac{\partial^2 V}{\partial r^2} + \frac{1}{r} \frac{\partial V}{\partial r} + \frac{1}{r^2} \frac{\partial^2 V}{\partial \theta^2} = 0$ with the following boundary conditions: (i) V is finite when $r \rightarrow 0$ (ii) $V = \sum C_n \cos n\theta$ on $r = a$.	CO2	K2

Python Programming (BCC-302)

Unit- 2

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1.	Write a Python program that checks whether a given number is positive, negative, or zero using if-elif-else.	CO2	K1
2.	What is the difference between break and continue in Python loops? Give an example.	CO2	K2
3.	How is the else block used with loops in Python? Write an example.	CO2	K2
4.	Write a while loop to print the first 5 multiples of 3.	CO2	K1
5.	What does the pass statement do in Python? Demonstrate with an example.	CO2	K2
	Questions (Long Answer Questions)		
6.	Write a Python program that takes a number from the user and checks whether it is divisible by 3, 5, or both using conditional blocks.	CO2	K2
7.	Write a program to find the sum of all integers between 1 and 100 that are divisible by 7 using a for loop and continue to skip non divisible numbers.	CO2	K2
8.	Write a program that repeatedly asks the user to enter a password until they type the correct one, using a while loop and break.	CO2	K1
9.	Using a dictionary of student names as keys and marks as values, write a program that prints only those students who scored more than 75.	CO2	K2
10.	Write a Python program that generates the Fibonacci sequence up to 50 using a while loop.	CO2	K2

Assignment- 3

Data Structure (BCS-301)

Unit- 3

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
Questions (Short Answer Questions)			
1	What do you understand by stable and in place sorting?	CO3	K2
2	Differentiate sequential search and binary search.	CO3	K3
3	Give example of one each stable and unstable sorting techniques.	CO3	K2
4.	How does bubble sort work? Explain.	CO3	K3
Questions (Long Answer Questions)			
5.	How binary search is different from linear search? Apply binary search to find item 40 in the sorted array: 11,22, 30, 33,40, 44,55, 60, 66,77, 80, 88, 99. Also discuss the complexity of binary search.	CO3	K3
6.	Explain any three commonly used hash function with the suitable example. The highest function H defined as $H(\text{key}) = \text{key} \bmod 7$, with linear probing, is used to insert the key 37,38, 72, 48, 98, 11, 66, into a table indexed from 0 to 6. What will be the location of key 11? Justify your answer, also count the total number of collisions in this probing.	CO3	K3
7.	Why is quick sort named as quick? Show the steps of quick sort on the following set of elements: 25,57, 48,37,12,92,86,33. Assume the first element of the list to be the pivot element.	CO3	K3
8.	What is Hashing? Explain division method to compute the hash function and also explain the collision resolution strategies used in hashing.	CO3	K3
9.	Write an algorithm for Heap Sort. Use Heap sort algorithm, sort the following sequence: 18, 25, 45, 34, 36, 51, 43, and 24.	CO3	K2
10.	Differentiate between linear and quadratic probing techniques.	CO3	K2
11.	Write an algorithm for merge sort and apply on following elements. 45,32, 65,76, 23,12, 54, 67, 22. 87.	CO3	K2

Discrete Structure and Theory of Logic (BCS-303) Unit- 3

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1.	Write the converse, inverse and contrapositive of a statement S is given as " If $x < 2$ then $x + 4 < 6$ ".	CO3	K2
2.	Show that $((P \vee Q) \wedge \neg(\neg Q \vee \neg R)) \vee (\neg P \vee \neg Q) \vee (\neg P \vee \neg R)$ is a tautology.	CO3	K3
3.	Prove that $(P \vee Q) \rightarrow (P \wedge Q)$ is logically equivalent to $P \leftrightarrow Q$.	CO3	K3
4.	Negate "All men are honest"	CO3	K3
	Questions (Long Answer Questions)		
5.	Prove the validity of the following argument:- If I get the job and work hard then I will get promoted. If I will get promoted, then I will be happy. I will not be happy. Therefore I will not get the job or I will not work hard.	CO3	K3
6.	Show that statements $P \rightarrow Q$ and $Q' \rightarrow P'$ are logically equivalent.	CO3	K3
7.	Rewrite the following arguments using quantifiers, variables and predicate symbols. i. All birds can fly. ii. Some men are genius. iii. Some numbers are not rational. iv. There is a student who likes mathematics but not geography.	CO3	K3
8.	Check whether below are Well Formed Formula or not i. $P \vee \wedge Q$ ii. $PQR \vee \wedge \neg \rightarrow$ iii. $\sim \sim \sim P$	CO3	K3
9.	Write any two Inference rule on your own for predicate logic.	CO3	K2
10.	All bird can fly. Is it a propositional logic statement? Explain in detail	CO3	K2

Computer Organization and Architecture (BCS-302)

Unit- 3

Note:- Attempt all questions in brief.

S.No.	Questions	COs	Bloom's Taxonomy
1.	How memory read and write operations are performed in the computer system.	CO3	K2
2.	What are the different types of instruction formats?	CO3	K2
3.	Represent the following conditional control statements by two register transfer statements with control functions. If(P=1)then(R1<- R2)else if(Q=1)then(R1<- R3)	CO3	K2
4.	Differentiate between horizontal and vertical microprogramming.	CO3	K2
5.	Write short note on RISC.	CO3	K2
6.	Explain the concept of pipelining and also explain the types of pipelining.	CO3	K2
7.	Write a program to evaluate the arithmetic statement. $P = ((X - Y + Z) * (A \wedge B)) / (C \wedge D * E)$ By using (1)Two address instructions(2)One address instruction(3) zero address instructions	CO3	K3
8.	List the differences between hardwire and micro programmed control in tabular format. Write the sequence of control steps for the following instruction for single bus architecture: $R1 \leftarrow R2 * (R3)$	CO3	K2
9.	Differentiate between horizontal and vertical microinstructions. A vertical micro programmed control unit support 512 instruction. The system is using 8 conditional flags and contain 31 control signals. Each instruction on an average required 1 micro operation. Calculate the approximate size of control memory in bytes.	CO3	K3
10	Explain hardwired control unit design using its block diagram. Explain generation of a control signals.	CO3	K2

Universal Human Values (BVE-301)

Unit- 3

Note:- Attempt all questions in brief.

SECTION –A (SHORT ANSWER TYPE QUESTIONS)		CO	Blooms Taxonomy Level
1	A	Bring out the difference between reverence and gratitude.	CO3 K1
	B	What do you understand by trust? Differentiate between intention and competence with examples.	CO3 K2
	C	What is Respect? When do we feel we have been disrespected?	CO3 K2
	D	Feeling of love lay down the basis of undivided society. Explain.	CO3 K2
	E	Bring out the contrast between excellence and competition. Can competition be destructive as well as constructive?	CO3 K1
SECTION –B (LONG ANSWER TYPE QUESTIONS)			
2	A	What are the programs needed to achieve the comprehensive human goal? List and define each briefly.	CO3 K1
	B	What is meaning of justice in human relationships? How does it follow from family to world family?	CO3 K2
	C	What do you mean by differentiation in relationship? What are the issues on which differentiation in relationship is prevalent in the society?	CO3 K2
	D	What values are necessary in human relationship? Explain each briefly.	CO3 K2
	E	What are the five dimensions of Human Endeavour/Order? How are they helpful in achieving the comprehensive human goal?	CO3 K2

Technical Communication (BAS-301)

Unit- 3

Note:- Attempt all questions in brief.

S.No.	Questions	CO	Bloom's Taxonomy
1.	What is thesis writing? Describe its structure and importance.	CO3	K2
2.	What is project report writing and what are the main parts that should be included to make it clear for the reader?	CO3	K1
3.	What is meant by Reference in technical article? How is it different from Bibliography? Explain with examples.	CO3	K2
4.	Write a proposal for receiving government grant for setting up a manufacturing unit for surgical appliances.	CO3	K1
5.	Discuss different types of reports along with the style of it.	CO3	K2
6.	Discuss the different types of pitching.	CO3	K2
7.	Discuss the structure of technical report in detail.	CO3	K1
8.	Draft a letter of transmittal (as a technical expert) for your proposal regarding the network system in your organization	CO3	K1
9.	“What is the best way to pitch an idea?” Support your answer with an example.	CO3	K2
10.	Discuss poise and illustration to make the presentation effective. or Why are passion, poise and the use of illustration important while speaking?	CO3	K2

Digital Electronics (BOE-310)

Unit- 3

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1.	What is the difference between characteristic and excitation table.	CO3	K1
2.	Differentiate between combinational and sequential circuits.	CO3	K1
3.	Give the difference between positive and negative edge triggering.	CO3	K1
	Questions (Long Answer Questions)		
4.	Explain the term registers.	CO3	K1
5.	Define bi-directional shift register. Draw and explain 3-bit bi-directional shift register using D flip-flop.	CO3	K2
6.	Describe the Design of J-K FF using T FF.	CO3	K2
7.	Design and implement MOD-10 synchronous counter.	CO3	K2
8.	Explain the working and circuit of a modulo-5 asynchronous counter using gates.	CO3	K2
9.	Design a 3-bit UP/DOWN counter with a direction control M, using JK flip-flops.	CO3	K2
10.	Discuss excitation table for SR, JK, T and D flip flop.	CO3	K2

Mathematics-IV (BAS-303)

Unit- 3

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's					
	Questions (Short Answer Questions)							
1.	The first four moments about mean of a frequency distribution are 0, 100, −7 and 35000. Discuss the kurtosis of the distribution.	CO3	K2					
2.	The first three central moments of a distribution are 0,15, −31. Find the moment coefficient of skewness.	CO3	K4					
3.	Find the relation between moment about the mean and moment about arbitrary point. The first four moments of a distribution about the value 4 of the variate are -1.5, 17, -30 and 108. Calculate the first four moments (i) about mean (ii) about origin (iii) about $x = 2$ and find β_1 and β_2 .	CO3	K2					
	Questions (Long Answer Questions)							
4.	The first four moments of a distribution about the value 4 of a variable are −1.5, 17, −30 and 108. Find the moments about mean, about origin ; β_1 and β_2 . Also, find the moments about the point $x = 2$.	CO3	K2					
5.	Find the least square line for the following points: $(-1, 0), (0, 9), (1, 7), (2, 5), (3, 4), (4, 3), (5, 0), (6, -1)$	CO3	K3					
6.	Establish the formula: $\sigma_{x-y}^2 = \sigma_x^2 + \sigma_y^2 - 2r_{xy}\sigma_x \sigma_y$ where r_{xy} is the correlation coefficient between x and y .	CO3	K5					
7.	If θ is acute angle between two regression lines in case of two variables x and y , then show that $\tan \theta = \frac{1-r^2}{r} \left(\frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \right)$ where the symbols have their usual meaning. Explain the significance of the formula when $r = 0, r = \pm 1$.	CO3	K3					
8.	The regression lines of y on x and x on y are: $y = ax + b$ and $x = cy + d$. Show that $\frac{\sigma_y}{\sigma_x} = \sqrt{\frac{a}{c}}$, $\bar{x} = \frac{bc+d}{1-ac}$ and $\bar{y} = \frac{ad+b}{1-ac}$	CO3	K2					
9.	In partially destroyed laboratory record of an analysis of correlation data, the following results are only legible: Variance of $x = 9$ Regression lines: $8x - 10y + 66 = 0$ and $40x - 18y = 214$. What are (A) Means of x and y (B) Standard deviation of y and (C) Coefficient of correlation between x and y	CO3	K5					
10.	A computer while calculating correlation coefficient between two variables x and y from 25 pairs of observations obtained the following results: <table><tr><td>$\sum x = 125$</td><td>$\sum y = 100$</td><td>$\sum x^2 = 650$</td><td>$\sum y^2 = 460$</td><td>$\sum xy = 508$</td></tr></table> It was, however, later discovered at the time of checking that he had copied down two pairs as (6,14), (8,6) while the correct values were (8,12), (6,8). Obtain the correct value of correlation coefficient between x and y .	$\sum x = 125$	$\sum y = 100$	$\sum x^2 = 650$	$\sum y^2 = 460$	$\sum xy = 508$	CO3	K5
$\sum x = 125$	$\sum y = 100$	$\sum x^2 = 650$	$\sum y^2 = 460$	$\sum xy = 508$				

Python Programming (BCC-302)

Unit- 3

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1.	What are immutable and mutable data types in Python? Show with code how a list can be modified but a tuple cannot.	CO3	K2
2.	What is the purpose of a dictionary in Python? Create a dictionary of 3 countries and their capitals, then print all keys and values.	CO3	K1
3.	What are string methods in Python? Use any two string methods (upper(), replace()) with examples.	CO3	K2
4.	Why do we use functions in Python programs? Write a function that takes a name as input and prints a welcome message.	CO3	K2
5.	What is the main difference between a list and a tuple? Create a tuple (1,2,3,4) and try to modify the second element. Explain the result.	CO3	K2
	Questions (Long Answer Questions)		
6.	Explain string manipulation in Python with examples. Write a program that counts vowels and consonants in a given string.	CO3	K2
7.	What are the advantages of using tuples in Python? Write a program that stores (student_name, roll_no, marks) in a tuple and prints all details.	CO3	K1
8.	What are built-in string, list, and dictionary functions in Python? Write a program that: • Converts "python programming" to uppercase • Sorts a list of numbers • Finds the length of a dictionary	CO3	K2
9.	Why are functions called “building blocks” of Python programs? Write a function factorial(n) that returns the factorial of a number using a loop.	CO3	K2
10.	Compare string, list, and dictionary manipulations in Python. Write one small program each that shows: • String reversal • Removing duplicates from a list • Updating a dictionary value	CO3	K2

Assignment- 4

Data Structure (BCS-301)

Unit- 4

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1	Discuss the concept of "successor" and "predecessor" in Binary Search Tree.	CO4	K2
2	Define extended binary tree, full binary tree, strictly binary tree and complete binary tree.	CO4	K3
3.	Write short notes on mini heap.	CO4	K3
	Questions (Long Answer Questions)		
4.	Construct a B-tree on following sequence of inputs. 10, 20, 30, 40, 50, 60, 70, 80, 90. Assume that the order of the B-tree is 3.	CO4	K3
5.	Explain Huffman algorithm. Construct Huffman tree for MAHARASHTRA, with its optimal code.	CO4	K3
6.	What is the difference between binary search tree and heap? For a given sequence of numbers, construct a heap and a BST. 34,23,67,45,12,54,87,43,98,75,84,93,31.	CO4	K3
7.	(a) Write the algorithm for deletion of an element in Binary search tree (b) Construct the binary tree for the following: In-order : Q,B,K,C,F,A,G,P,E,D,H,R Preorder: G,B,Q,A,C,K,F,P,D,E,R,H find the post order of the tree.	CO4	K3
8.	(i) Why does time complexity of search operation in B-Tree is better than Binary Search Tree (BST)? (ii) Insert the following keys into an initially empty B-tree of order a, g, f, b, k, d, h, m, j, e, s, i, r, x, c, l, n, t, u, p (iii) What will be the resultant B-Tree after deleting keys j, t and d in sequence?	CO4	K2
9.	(a) Draw a binary tree with following traversals: Inorder: B C A E G D H F I J Preorder: A B C D E G F H I J (b) Consider the following AVL Tree and insert 2, 12, 7 and 10 as new node. Show proper rotation to maintain the tree as AVL.	CO4	K3
10.	Define tree, binary tree, complete binary tree and full binary tree. Write algorithms or function to obtain traversals of a binary tree in preorder, postorder and inorder.	CO4	K2

Discrete Structure and Theory of Logic (BCS-303) Unit- 4

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1	Show that every cyclic group is abelian.	CO4	K2
2	Write all subgroups of $(\mathbb{Z}_4, +_4)$.	CO4	K3
3.	What is Ring? Define elementary properties of Ring with example.	CO4	K3
	Questions (Long Answer Questions)	CO4	
4.	Consider the group $G = \{1, 2, 3, 4, 5, 6\}$ under multiplication modulo 7. Check whether it is Abelian Group or not?	CO4	K3
5.	What do you mean by cosets of a subgroup? Consider the group $\mathbb{Z}$ of integers under addition and the subgroup $H = \{\dots, -12, -6, 0, 6, 12, \dots\}$ considering of multiple of 6. Find the cosets of H in $\mathbb{Z}$	CO4	K3
6.	How many subgroups are possible for group $(\mathbb{Z}_{12}, +_{12})$?	CO4	K3
7.	State and prove Lagrange's theorem for group.	CO4	K3
8.	Prove that every cyclic group is an abelian group.	CO4	K2
9.	Prove that the set $S = \{0, 1, 2, 3\}$ forms a ring under addition and multiplication modulo 4 but not a field.	CO4	K3
10.	Define the following with suitable example: (i) Cyclic group (ii) Zero divisor of ring.	CO4	K2
11.	Let $G = \{1, -1, i, -i\}$ be the multiplicative group, where $i = \sqrt{-1}$ i. Determine whether G is an abelian. ii. If G is a cyclic group, then determine generator of G .	CO4	K2

Computer Organization and Architecture (BCS-302)

Unit- 4

Note:- Attempt all questions in brief.

S.No.	Questions	COs	Bloom's Taxonomy
1.	Define HIT and MISS ratio in memory with an example.	CO4	K2
2.	List the difference between static RAM and dynamic RAM.	CO4	K2
3.	Define virtual memory.	CO4	K2
4.	What do you mean by page fault?	CO4	K2
5.	What is the difference between 2D and 2.5D memory organization.	CO4	K2
6.	Consider a cache consisting of 256 blocks of 16 words each for a total of 2096 words and assume that the main memory is addressable by a 16 bits address and it consists of 4K block. How many bits are there in each of TAG, SET, WORD field for 2-way set associative technique,	CO4	K2
7.	Discuss the Memory Hierarchy in computer system with regard to Speed, Size, and Cost.	CO4	K3
8.	A digital computer has a memory unit of 64K X 16 and cache memory of 1 K words. The cache uses direct mapping with a block size of four words. 1. How many bits are there in tag, index, block, and word field of the address format? How many bits are there in each word of cache, and how they are divided into functions? Include a valid bit.	CO4	K2
9.	How is the Virtual address mapped into physical address? What are the different methods of writing into cache?	CO4	K3
10	Explain cache performance measure. Evaluate the performance of 2-level cache memory.	CO4	K2

Universal Human Values (BVE-301)

Unit- 4

Note:- Attempt all questions in brief.

SECTION –A (SHORT ANSWER TYPE QUESTIONS)			CO	Blooms Taxonomy Level
1	A	Define harmony in nature	CO4	K1
	B	What do you mean by co-existence?	CO4	K2
	C	Define the terms Gathansheel, Gathanpurna, Kriyapurnata and Acharanpurnata.	CO4	K2
	D	What do you mean by ‘innateness’? What is the innateness in the four orders?	CO4	K2
	E	"As the seed, thus the plant". Explain.	CO4	K1
SECTION –B (LONG ANSWER TYPE QUESTIONS)				
2	A	There are four orders in nature. How does each order participate in the harmony in the nature?	CO4	K1
	B	Explain how there is recyclability and self-regulation in nature.	CO4	K2
	C	Comment on the statement: “Nature is limited and space is unlimited.”	CO4	K2
	D	Explain the concept of holistic perception of harmony in existence.	CO4	K2
	E	Outline the differences and similarities between animal order and human order.	CO4	K2

Technical Communication (BAS-301)

Unit- 4

Note:- Attempt all questions in brief.

S.No	Questions	CO	Bloom's Taxonomy
1.	Differentiate between tact and intelligence.	CO4	K2
2.	What is emotional intelligence and why is it important in professional life?	CO4	K1
3.	Discuss how communication and social competence are interrelated?	CO4	K2
4.	Discuss the role of feeling and context in social competence.	CO4	K1
5.	'Providing and receiving the feedback can enhance communication and social competence.' Illustrate the statement with suitable example.	CO4	K2
6.	What are the main types of thinking skills?	CO4	K2
7.	Discuss the different types /style of leadership and identify their role according to their style .	CO4	K1
8.	Discuss the strategies to improve leadership communication.	CO4	K1
9.	What is leadership communication? Why should a leader be socially competent to be a good leader?	CO4	K2
10.	Discuss the challenges of leadership communication with the help of example.	CO4	K2

Digital Electronics (BOE-310)

Unit- 4

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1.	Define race around condition in JK flip flop.	CO4	K1
2.	Define term propagation delay.	CO4	K1
3.	Define the use of clock in digital circuits.	CO4	K1
	Questions (Long Answer Questions)		
4.	Define critical race and non-critical race.	CO4	K1
5.	Discuss Mealy and Moore finite state machine with an example.	CO4	K2
6.	Illustrate the State reduction technique for Digital Circuits.	CO4	K2
7.	Design a sequential circuit with two flip flops, A & B and one input X. When X=0 state of the circuit remains the same, when X =1 circuit passes through the state transition from 00 to 01 to 11 to 10 back to 00 And repeat.	CO4	K3
8.	A sequential circuit has two J K flip flops A & B, two inputs X & Y, and one output Z. The equations defining this system are as following: $JA = BX + B'Y'$ $KA = B'XY'$ $JB = A'X$ $KB = A + XY'$ $Z = AXY + BX'Y'$ Design the circuit.	CO4	K3
9.	Explain the term, Hazard. Define different types of Hazards along with detection and reduction of Hazards.	CO4	K2
10.	Design a primitive state diagram and state table for a circuit with two asynchronous inputs (X and Y) and one output Z. This circuit is to be designed so that if any change takes place on X and Y, Z is to change states. Assume initially that the two inputs never change Simultaneously.	CO4	K2

Mathematics-IV (BAS-303)

Unit- 4

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
Questions (Short Answer Questions)			
1.	Find the probability of getting 4 heads in 6 tosses of a fair coin.	CO4	K2
2.	Comment on the following statement: For a binomial distribution mean is 6 and variance is 9.	CO4	K2
3.	Using Poisson distribution, find the probability that the ace of the spades will be drawn from a pack of well – shuffled cards at least once in 104 consecutive trials.	CO4	K1
Questions (Long Answer Questions)			
4.	What is total probability theorem?	CO4	K1
5.	Two urns contain 4 white, 6 blue and 4 white, 5 blue balls respectively. One of the urns is selected at random and a ball is drawn from it. If the ball drawn is white, find the probability that it is drawn from (i) first urn (ii) second urn.	CO4	K3
6.	Find the moment generating function of Binomial distribution and hence find its mean and variance.	CO4	K4
7.	If sum and product of the mean and variance of a binomial distribution are $\frac{15}{3}$ and $\frac{10}{3}$ respectively. Find the distribution.	CO4	K4
8.	In a certain factory turning out razor blades, there is a small chance of 0.002 for any blade to be defective. The blades are supplied in packets of 10. Calculate the approximate number of packets containing no defective, one defective and two defective blades in a consignment of 10,000 packets. (Given: $e^{-0.02} = 0.9802$)	CO4	K3
9.	In a sample of 1000 cases, the mean of certain test is 14 and standard deviation is 2.5. Assuming the distribution to be normal, find a) How many students score between 12 and 15? b) How many score above 18? c) How many score below 8? d) How many score 16?	CO4	K4
10.	In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and standard deviation of the distribution. It is given that if $f(t) = \frac{1}{\sqrt{2\pi}} \int_0^t e^{-\frac{1}{2}x^2} dx$ then $f(0.5) = 0.19$ and $f(1.4) = 0.42$.	CO4	K4

Python Programming (BCC-302)

Unit- 4

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1.	What is the difference between read(), readline(), and readlines() in Python? Give examples.	CO4	K2
2.	Write a program that writes three lines into a text file and then reads them back using readlines().	CO4	K1
3.	Explain the purpose of the file pointer. Show with code how seek() can be used to move the pointer to the beginning of a file.	CO4	K2
4.	How is writelines() different from write() in Python? Demonstrate with code.	CO4	K2
5.	Write a program that reads only the first line of a file using readline().	CO4	K2
	Questions (Long Answer Questions)		
6.	Explain different file modes (r, w, a, r+, w+, a+) in Python with examples. Write a program to demonstrate the use of at least two modes	CO4	K2
7.	Write a program to copy the contents of one text file into another text file.	CO4	K2
8.	Explain the use of seek() and tell() functions in Python. Write a program that moves the file pointer to the 10th character and reads from there till the end of file.	CO4	K2
9.	Create a file containing multiple numbers (one per line). Write a program to read the file, calculate the sum of all numbers, and write the result into another file.	CO4	K1
10.	Write a program that writes a list of strings into a file using writelines(). Then, reopen the file and display its contents line by line using a loop.	CO4	K2

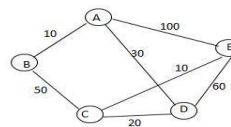
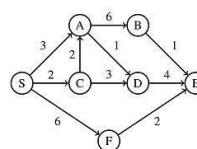
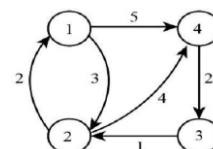
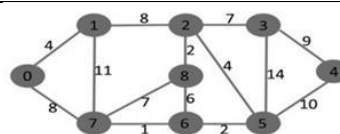
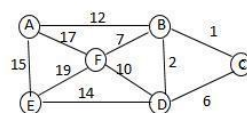
Assignment- 5

Data Structure (BCS-301)

Unit- 5

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1	List the different types of representation of graphs.	CO5	K2
2	What is minimum cost spanning tree? Give its applications.	CO5	K3
3.	Write different representations of graphs in the memory.	CO5	K3
	Questions (Long Answer Questions)		
4.	Define spanning tree. Also construct minimum spanning tree using prim's algorithm for the given graph.	CO5	K3
5.	Use Dijkstra's algorithm to find the shortest paths from source to all other vertices in the following graph.	CO5	K3
6.	write the Floyd Warshall algorithm to compute the all pairs shortest path. Apply the algorithm on following graph.	CO5	K3
7.	Explain the detail about the graph traversal technique with suitable examples.	CO5	K3
8.	Differentiate between DFS and BFS. Draw the breadth First Tree for the above graph.	CO5	K2
9.	Find the minimum spanning tree in the following graph using Kruskal's algorithm.	CO5	K3
10.	Describe Dijkstra's algorithm for finding shortest path. Describe its working for the graph given below.	CO5	K3



Discrete Structure and Theory of Logic (BCS-303) Unit- 5

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1	Show that there does not exist a graph with 5 vertices with degrees 1, 3, 4, 2, 3 respectively.	CO5	K2
2	What is difference between Full Binary Tree and Complete Binary Tree? Explain it with examples.	CO5	K3
3.	Define Pigeon-hole principle.	CO5	K3
	Questions (Long Answer Questions)		
4.	Difference between Tree and Graph.	CO5	K3
5.	Find the adjacency matrix $A = [a_{ij}]$ of graph given in following figure:- 	CO5	K3
6.	Draw 2 planar and two non planar graph.	CO5	K3
7.	Draw 2 Bipartite Graphs and perform graph coloring on them.	CO5	K3
8.	Define Isomorphic Graph and Hamiltonian Graph.	CO5	K2
9.	Explain Euler's formula. Determine number of regions if a planar graph has 30 vertices of degree 3 each.	CO5	K2
10.	Show that $K_{3,3}$ satisfies in equality $ E \leq 3 V - 6$, but it is nonplanar. (V=No. of Vertices, E=No. of Edges, R=No. of Regions)	CO5	K3

Computer Organization and Architecture (BCS-302)

Unit- 5

Note:- Attempt all questions in brief.

S.No.	Questions	COs	Bloom's Taxonomy
1.	List down the functions performed by an Input /Output unit.	CO5	K2
2.	Why does the DMA get priority over the CPU when both request memory transfer?	CO5	K2
3.	Give the block diagram of DMA controller. Why are the read and write control lines in a DMA controller bidirectional?	CO5	K2
4.	Write the difference between serial and parallel communication.	CO5	K2
5.	What is I/O control method?	CO5	K2
6.	Explain the difference between vectored and non-vectored interrupt. Explain stating examples of each.	CO5	K2
7.	Give the block diagram of DMA controller. Why are the read and write control lines in a DMA controller bidirectional?	CO5	K3
8.	What do you mean by asynchronous data transfer? Explain strobe control and hand shaking mechanism.	CO5	K2
9.	Write down the differences between isolated and memory mapped I/O. Also discuss advantages and disadvantages of isolated I/O and memory mapped I/O.	CO5	K3
10	Define interrupt. Also discuss different types of interrupts.	CO5	K2

Universal Human Values (BVE-301)

Unit- 5

Note:- Attempt all questions in brief.

SECTION –A (SHORT ANSWER TYPE QUESTIONS)			CO	Blooms Taxonomy Level
1	A	What do you mean by professional ethics?	CO5	K1
	B	What is utility value and artistic value? How are both important in human life? Explain with example.	CO5	K2
	C	What do you mean by ‘universal human order’?	CO5	K2
	D	Comment on Profession – in the light of comprehensive human goal.	CO5	K2
	E	What are the implications of value based living at all four levels of living? Explain.	CO5	K1
SECTION –B (LONG ANSWER TYPE QUESTIONS)				
2	A	What is ethical human conduct? Explain in terms of values, policies, and character with appropriate examples.	CO5	K1
	B	How do the current world views lead to contradictions and dilemmas in professional life? – Explain.	CO5	K2
	C	Describe briefly the criteria for evaluation of holistic technology. Support your answer with an example.	CO5	K2
	D	Mention some of the unethical practices in society today. How do the prevailing world views lead to such unethical practices?	CO5	K2
	E	What do you mean by competence in professional ethics? Elaborate with examples.	CO5	K2

Technical Communication (BAS-301)

Unit- 5

Note:- Attempt all questions in brief.

S. No.	Questions	CO	Bloom's Taxonomy
1.	Discuss speech and personality.	CO5	K2
2.	What are the different types of personality and how do they affect the way people think, feel and behave in daily life and communication.	CO5	K1
3.	Discuss the Maslow theory of personality.	CO5	K2
4.	What are the possible harmful effects of content creation on social media? Discuss its impact on a person's mental health, privacy and social life.	CO5	K1
5.	Discuss the ethics to handle social media effectively.	CO5	K2
6.	Discuss the different strategies to create content for blogs.	CO5	K2
7.	Enlist the different content creation idea.	CO5	K1
8.	Discuss the tips for building your email list as a content creator.	CO5	K1
9.	Write short note on the following: Webinars, Podcasts, blogs.	CO5	K2
10.	How does Swami Vivekananda define personality? Explain.	CO5	K2

Digital Electronics (BOE-310)

Unit- 5

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1.	How many address lines are needed to represent 8K memory?	CO5	K1
2.	Elaborate the term Fan-in in digital circuits.	CO5	K1
3.	Why is ECL logic faster than TTL?	CO5	K1
	Questions (Long Answer Questions)		
4.	Compare static RAM and dynamic RAM.	CO5	K1
5.	Define the TTL (Transistor-Transistor-Logic) logic Family used for digital circuits.	CO5	K2
6.	Design a BCD to Excess-3 code converter and implement it using a suitable PLA.	CO5	K2
7.	Describe the difference between PAL & PLA using neat diagram and suitable examples.	CO5	K2
8.	Define the SRAM cell with working and circuit diagram along with applications.	CO5	K2
9.	Define noise margin, Fan-out as characteristics of logic families. Implement NAND gate with CMOS.	CO5	K2
10.	Design 8K x 8 RAM memory system, using 1Kx8 memory ICs.	CO5	K2

Mathematics-IV (BAS-303)

Unit- 5

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's																																
	Questions (Short Answer Questions)																																		
1.	What is the number of degree of freedom for a contingency table of order 5×4 ?	CO5	K2																																
2.	Write short note on t-test.	CO5	K2																																
3.	While testing the significance of difference of two sample means in case of small samples, how is degree of freedom calculated?	CO5	K1																																
	Questions (Long Answer Questions)																																		
4.	What do you mean by statistical quality control?	CO5	K1																																
5.	A random sample of size 16 values from a normal population showed a mean of 53 and sum of square of deviations from the mean is 150. Can this sample be regarded as taken from the population having 56 as mean? Obtain 95% and 99% confidence limits of the mean population.	CO5	K1																																
6.	A group of 5 patients treated with the medicine A weigh 42, 39, 48, 60 and 41 kgs. A second group of 7 patients from the same hospital treated with medicine B weigh 38, 42, 56, 64, 68, 69 and 62 kgs. Do you agree with the claim that medicine B increases the weigh significantly? It is given that the value of t at 10% level of significance for 10 dof is 1.81.	CO5	K2,K ₂																																
7.	Out of 8000 graduates in a town, 800 are female, out of 1600 graduate employees, 120 are females. Use χ^2 test to determine if any distinction is made in appointment on the basis of sex. (Given that the value of χ^2 for 1 degree of freedom at 5% level of significance is 3.841.)	CO5	K1																																
8.	The following figure shows the number of defectives in 15 samples, each sample containing 1500 items. <table><tr><th>Sample No.</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th><th>9</th><th>10</th><th>11</th><th>12</th><th>13</th><th>14</th><th>15</th></tr><tr><td>Number of defectives</td><td>150</td><td>175</td><td>225</td><td>180</td><td>75</td><td>125</td><td>200</td><td>325</td><td>105</td><td>120</td><td>165</td><td>100</td><td>80</td><td>90</td><td>120</td></tr></table> Calculate the values for control line and control limits for p – chart, np – chart and C – chart. Draw the p – chart, np – chart and C – chart and comment is the process is under control or not.	Sample No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Number of defectives	150	175	225	180	75	125	200	325	105	120	165	100	80	90	120	CO5	K1
Sample No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15																				
Number of defectives	150	175	225	180	75	125	200	325	105	120	165	100	80	90	120																				
9.	The theory predicts the proportion of beans in the four groups, G_1, G_2, G_3, G_4 should be in the ratio 9:3:3:1. In an experiment with 1600 beans the numbers in the four groups were 882, 313, 287 and 118. Does the experimental result support the theory? (Given that $\chi^2 = 7.815$ at 5% level of significance for degree of freedom = 3)	CO5	K2																																

Python Programming (BCC-302)

Unit- 5

Note:- Attempt all questions in brief.

S. No.	Questions	COs	Bloom's
	Questions (Short Answer Questions)		
1.	What is the use of NumPy in Python? Write a program to create a NumPy array of 10 numbers and find their mean.	CO5	K1
2.	How does matplotlib.pyplot.plot() work? Write code to plot a simple line graph of numbers 1 to 5.	CO5	K2
3.	Write a program in Pandas to read a CSV file named "data.csv" and display the first 5 rows.	CO5	K1
4.	Write a program using Tkinter that contains a Button widget, and when clicked, displays "Hello, Tkinter!"	CO5	K2
5.	What are Python IDEs used for? List two IDEs and explain how they help in writing Python code.	CO5	K1
	Questions (Long Answer Questions)		
6.	Explain the role of NumPy in scientific computing. Write a program that creates a 3x3 NumPy matrix and finds its transpose and determinant.	CO5	K1
7.	Write a program using Pandas to create a DataFrame of 5 employees (Name, Department, Salary). Display the maximum, minimum, and average salary.	CO5	K2
8.	Write a Pandas program to read a CSV file, filter rows where marks are greater than 50, and save the filtered data into a new file.	CO5	K1
9.	Explain Tkinter widgets with examples. Write a program to create a Tkinter GUI that contains: • A Label with the text "Enter your name" • An Entry widget to accept input • A Button that displays the entered name in a Label when clicked	CO5	K2
10.	Create a Tkinter program with two buttons: Add and Subtract. When clicked, they should perform addition and subtraction of two numbers entered in Entry widgets.	CO5	K2